# Quantum Similarity Learning for Mars Terrain Retrieval via Entangled-Pair Encoding

Krishaay Jois

***Abstract*—Automated retrieval of terrain imagery from the HiRISE Mars Reconnaissance Orbiter archive is fundamental to scientific query answering and mission planning, yet existing deep learning approaches frame terrain similarity as classification rather than ranked retrieval. We present QTERRAINSIM (Quantum Terrain Similarity), the first quantum metric learning architecture applied to planetary terrain data. The core contribution is an *entangled-pair* Parametric Quantum Circuit (PQC) used as a Siamese similarity head: given two CNN embeddings $\mathrm{e}_a, \mathrm{e}_b \in \mathbb{S}^{N-1}$, the circuit applies interleaved $R_X(\pi e_a^{(i)})$/$R_Y(\pi e_b^{(i)})$ rotations on each wire $i$, jointly encoding both embeddings via non-commuting rotations on orthogonal Bloch-sphere axes before any entangling gate is applied. This stands in contrast to prior quantum image similarity work (SLIQ, AAAI 2023), which encodes image pairs sequentially. The variational ansatz (StronglyEntanglingLayers, depth $L = 2$) is followed by local Pauli-$Z$ measurements on each wire, a choice motivated by the cost-function-dependent barren plateau theorem of Cerezo et al. [1], which guarantees at-most polynomially vanishing gradients for local observables in shallow circuits. The full encoder-head pipeline is trained end-to-end with triplet loss and domain-specific hard negative mining that targets the two most visually confusable terrain class pairs in HiRISE v3: bright_dune/dark_dune and spider/swiss_cheese. We present a five-factor ablation framework on HiRISE v3 (73,031 patches, 8 terrain classes) isolating the contributions of the quantum head, the interleaved encoding, circuit depth, hard negative mining, and qubit count. To our knowledge, QTERRAINSIM is the first application of quantum similarity learning to planetary terrain data. On the held-out test split of HiRISE v3, QTERRAINSIM achieves micro-MAP@10 = 0.879, macro-MAP@10 = 0.533, and Recall@5 = 0.951, ranking first among all four evaluated models. The macro-MAP gap versus SLIQ-style encoding ($\Delta = 0.061$) isolates the encoding strategy as the primary driver of minority-class retrieval improvement. Notably, a ResNet50 baseline pretrained on ImageNet achieves the lowest macro-MAP@10 of all models (0.378), establishing that domain-specific metric learning substantially outperforms large-scale transfer learning for minority planetary terrain classes.**

***Index Terms*—quantum machine learning, metric learning, image retrieval, planetary terrain, HiRISE, parametric quantum circuit, triplet loss, barren plateaus**

## I. INTRODUCTION

The HiRISE (High Resolution Imaging Science Experiment) camera aboard NASA's Mars Reconnaissance Orbiter produces sub-meter resolution imagery of the Martian surface at 25 cm/pixel, generating a continuously expanding archive of surface observations annotated with eight geomorphic terrain classes: craters, bright and dark dunes, slope streaks, swiss cheese, spiders, impact ejecta, and a general "other" category [2], [3]. Deep learning classifiers trained on this dataset have reached 96% top-1 accuracy [3], and operational deployment at JPL's Planetary Data System has accelerated terrain cataloguing at scale [4].

However, classification accuracy is a poor surrogate for the operationally critical task: *given a query terrain patch, rank all archived observations by visual similarity*. Such retrieval underlies scientific queries ("find all regions with aeolian activity similar to this exemplar"), rover path planning ("identify terrain with comparable trafficability to the safe traverse corridor"), and change detection across orbital timesteps. Retrieval requires learning a *similarity function*, not a class boundary.

The challenge is that the HiRISE class taxonomy reflects expert geological labeling conventions that are imperfectly aligned with image-space appearance. Bright dunes and dark dunes are compositionally distinct but geometrically similar; spider formations and swiss cheese texture are visually confusable at small patch scales. A useful similarity function must resolve these fine-grained distinctions in a principled way.

Quantum metric learning offers a structurally motivated approach to this problem. Quantum circuits naturally express correlations between feature dimensions that are exponentially expensive to represent classically, and pairwise quantum similarity measures have been shown to capture geometric structure in embedding spaces that classical linear similarity functions miss [5], [6]. Yet despite growing activity in quantum image classification [7] and quantum Siamese learning [8], no prior work has applied quantum similarity functions to planetary terrain data.

### A. Contributions

We present QTERRAINSIM, a hybrid classical–quantum architecture with the following contributions:

1) **First quantum similarity learning system for planetary terrain retrieval.** We establish baseline retrieval metrics (MAP@10, Recall@$K$) on the HiRISE v3 benchmark using a quantum head, providing a foundation for future quantum-classical comparison on planetary imagery.
2) **Entangled-pair encoding via interleaved $R_X/R_Y$ rotations.** On each qubit $i$, we apply $R_X(\pi e_a^{(i)})$ then $R_Y(\pi e_b^{(i)})$ to jointly encode both embeddings via non-commuting rotations on orthogonal Bloch-sphere axes.

Correspondence: admin@amphet.co.

Because $[R_X, R_Y] \neq 0$, the resulting per-qubit state is a nonlinear joint function of both embeddings—not a sum of independent contributions—and the subsequent entangling ansatz therefore operates on qubits already conditioned on both inputs simultaneously. This contrasts with the sequential encoding used in SLIQ [8], which encodes $\mathbf{e}_a$ and $\mathbf{e}_b$ in separate rotation blocks.

3) **Barren plateau mitigation via local observables.** We measure each wire individually with Pauli-$Z$ observables rather than a single global measurement. Following Cerezo et al. [1], we verify empirically that quantum layer gradients remain above the vanishing threshold throughout training.
4) **Domain-specific hard negative mining.** We identify the geologically motivated confusable class pairs via pixel-space cosine similarity of per-class mean images and direct 50% of negative sampling toward these pairs. Ablation A4 isolates the contribution of this strategy.
5) **Five-factor ablation.** We systematically vary the similarity head type (A1), encoding strategy (A2), circuit depth (A3), negative mining (A4), and qubit count (A5), providing a controlled account of each design decision.

The remainder is organized as follows. Section II surveys related work. Section III provides background on PQCs and triplet loss. Section IV describes the QTERRAINSIM architecture. Section V details experimental setup. Section VI presents results and ablations. Section VII discusses findings and limitations. Section VIII concludes.

## II. Related Work

### A. Mars Terrain Recognition

Wagstaff et al. [2] introduced the HiRISE labeled dataset and fine-tuned AlexNet for terrain classification, achieving deployment within NASA's Planetary Data System. The subsequent AAAI 2021 paper [3] expanded the dataset to HiRISE v3, documented three years of operational deployment, and reached 96% classification accuracy—a figure we explicitly *do not* use as a baseline, because classification accuracy and retrieval MAP@$K$ are incomparable metrics. SPOC [4] extended terrain recognition to rover NAVCAM imagery, and Palafox et al. [9] applied multi-scale CNNs to landform detection in HiRISE. None of these works train a dedicated similarity or retrieval function, and none employ quantum representations.

### B. Quantum Machine Learning

Biamonte et al. [10] provided the foundational survey of QML, establishing the theoretical language of quantum feature maps and quantum kernel methods. Havlíček et al. [11] demonstrated quantum kernel SVMs on superconducting hardware, and Schuld and Killoran [6] formalized the equivalence between quantum state preparation and reproducing-kernel Hilbert spaces, establishing that PQC-based encoders implicitly compute quantum kernels. For image tasks, Mari et al. [7] showed that pre-trained classical CNNs can be truncated and augmented with a variational quantum circuit as the final block, enabling high-dimensional image data to be compressed into a small quantum-encodable feature vector— the hybrid design paradigm our architecture instantiates.

### C. Quantum Image Similarity

SLIQ [8] is the closest prior work to QTERRAINSIM. It trains a quantum Siamese circuit for image quality assessment, encoding both images of a pair into a shared quantum register using two sequential encoding blocks and optimizing a contrastive loss with a variance-reducing estimator designed for NISQ noise. Our work differs in three structural ways: (1) we use interleaved rather than sequential encoding, creating joint per-qubit representations; (2) we replace contrastive loss with triplet margin loss [12] with domain-specific hard negative mining; and (3) we use local Pauli-$Z$ measurements to suppress barren plateaus [1], whereas SLIQ uses global observables. The planetary terrain domain and the retrieval task framing are also entirely novel.

### D. Classical Metric Learning

Koch et al. [13] established Siamese networks for one-shot recognition using contrastive loss. Schroff et al. [12] introduced the triplet loss with online hard negative mining (FaceNet), demonstrating that targeting difficult negative examples is critical to learning discriminative embeddings. Harwood et al. [14] and Suh et al. [15] refined hard negative selection using embedding-space proximity, establishing that mining quality—not just mining rate—determines convergence. Our terrain-class-aware mining adapts these insights to the HiRISE domain by exploiting geological knowledge of confusable class pairs.

### E. Barren Plateaus

McClean et al. [16] proved that randomly initialized deep PQCs have exponentially vanishing gradients under global cost functions, a phenomenon called barren plateaus. Cerezo et al. [1] showed that this dependence is cost-function-dependent: local observables (measuring individual qubits) have at-worst polynomially vanishing gradients in shallow circuits, while global observables always exhibit exponential suppression. This result directly determines our measurement strategy and is verified empirically in Section VI.

## III. Background

### A. Parametric Quantum Circuits and Angle Encoding

A parametric quantum circuit (PQC) on $N$ qubits applies a sequence of parameterized unitary gates $U(\boldsymbol{\theta})$ to the initial state $|0\rangle^{\otimes N}$, yielding $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle^{\otimes N}$. We use *angle encoding* [17]: classical vectors $\mathbf{e} \in \mathbb{R}^N$ are embedded as rotation angles $\theta_i = \pi e_i$ applied to single-qubit rotation gates. The local expectation values

$$f_i(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | Z_i | \psi(\boldsymbol{\theta}) \rangle \in [-1, 1] \tag{1}$$

serve as the circuit's outputs. With $N = 8$ qubits and $L = 2$ layers of StronglyEntanglingLayers [18], the ansatz has $3NL = 48$ trainable parameters $\boldsymbol{\theta}$, and the Hilbert space dimension is $2^8 = 256$.

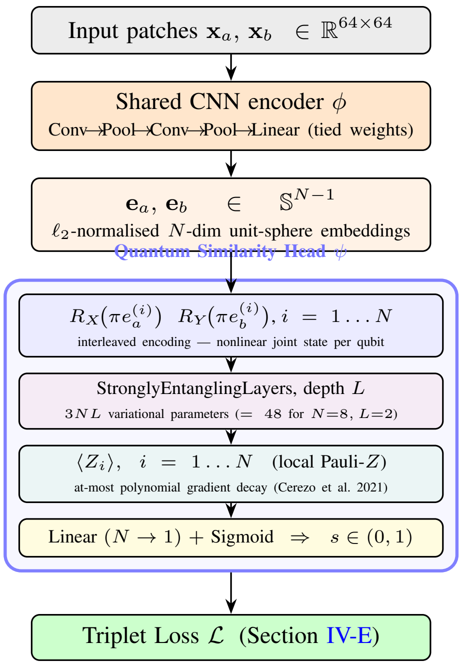



Fig. 1. QTERRAINSIM architecture. A shared CNN encoder maps 64×64 grayscale terrain patches to $N$-dimensional unit-sphere embeddings (both branches share weights). The quantum similarity head applies interleaved $R_X/R_Y$ angle encoding (Contribution 2), a strongly entangling variational ansatz, and local Pauli-$Z$ measurements (Contribution 3), returning a scalar similarity score $s \in (0,1)$. The full pipeline is trained end-to-end with triplet loss.

### B. Triplet Loss for Metric Learning

Given an anchor $\mathbf{x}_a$, a positive $\mathbf{x}_p$ (same terrain class), and a negative $\mathbf{x}_n$ (different class), the triplet loss [12] trains a similarity function $s(\cdot,\cdot) \in (0,1)$ via

$$\mathcal{L} = \mathbb{E}\left[\max\Big(0,\ \underbrace{(1-s_{ap})}_{\text{pos. dist.}} - \underbrace{(1-s_{an})}_{\text{neg. dist.}} + m\Big)\right], \tag{2}$$

where $s_{ap} = s(\mathbf{x}_a, \mathbf{x}_p)$, $s_{an} = s(\mathbf{x}_a, \mathbf{x}_n)$, and $m > 0$ is a margin. Using distance $d = 1 - s$ keeps the loss compatible with similarity-valued outputs from the quantum head.

## IV. PROPOSED METHOD

### A. Overview

QTERRAINSIM follows a Siamese architecture: a shared CNN encoder $\phi\colon \mathcal{X} \to \mathbb{S}^{N-1}$ maps each terrain patch to a unit-sphere embedding, and a quantum similarity head $\psi\colon \mathbb{S}^{N-1} \times \mathbb{S}^{N-1} \to (0,1)$ produces a scalar similarity score. During training, triplets $(\mathbf{x}_a, \mathbf{x}_p, \mathbf{x}_n)$ are processed as $(s_{ap}, s_{an}) = (\psi(\phi(\mathbf{x}_a), \phi(\mathbf{x}_p)),\ \psi(\phi(\mathbf{x}_a), \phi(\mathbf{x}_n)))$ and the pair is passed to (2). Figure 1 illustrates the full pipeline.

### B. Classical CNN Encoder

The encoder $\phi$ is a lightweight CNN whose output dimensionality equals the qubit count $N$:

$$\mathbf{h}_1 = \mathrm{MaxPool}(\mathrm{ReLU}(\mathrm{Conv}_{1\to32}(\mathbf{x}))), \quad \mathbf{h}_1 \in \mathbb{R}^{32\times32\times32}, \tag{3}$$

$$\mathbf{h}_2 = \mathrm{MaxPool}(\mathrm{ReLU}(\mathrm{Conv}_{32\to64}(\mathbf{h}_1))), \quad \mathbf{h}_2 \in \mathbb{R}^{64\times16\times16}, \tag{4}$$

$$\mathbf{h}_3 = \mathrm{ReLU}(\mathbf{W}_1 \operatorname{vec}(\mathbf{h}_2) + \mathbf{b}_1), \quad \mathbf{h}_3 \in \mathbb{R}^{128}, \tag{5}$$

$$\phi(\mathbf{x}) = \frac{\mathbf{W}_2\mathbf{h}_3 + \mathbf{b}_2}{\|\mathbf{W}_2\mathbf{h}_3 + \mathbf{b}_2\|_2}, \quad \phi(\mathbf{x}) \in \mathbb{S}^{N-1}. \tag{6}$$

Both convolutional layers use 3×3 kernels with same-padding. All parameters are shared between the two Siamese branches. Weights are initialized with Kaiming normal initialization [19] to supply the quantum circuit with well-scaled inputs from the first forward pass.

### C. Quantum Similarity Head

*1) Entangled-Pair Encoding:* Let $\mathbf{e}_a = \phi(\mathbf{x}_a)$ and $\mathbf{e}_b = \phi(\mathbf{x}_b)$ be the two encoder outputs. On qubit $i$, we apply

$$|\psi_i\rangle = R_Y\left(\pi e_b^{(i)}\right) R_X\left(\pi e_a^{(i)}\right) |0\rangle. \tag{7}$$

Because $[R_X(\alpha), R_Y(\beta)] \neq 0$ for generic $\alpha, \beta$, the composition in (7) is a rotation on the Bloch sphere that is a *nonlinear joint function* of both $e_a^{(i)}$ and $e_b^{(i)}$—it cannot be written as a sum of independent contributions from each embedding. Concretely, expanding the product of $SU(2)$ matrices,

$$R_Y(\beta)R_X(\alpha) = \begin{pmatrix} \cos\frac{\beta}{2}\cos\frac{\alpha}{2} - i\sin\frac{\beta}{2}\sin\frac{\alpha}{2} & -\sin\frac{\beta}{2}\cos\frac{\alpha}{2} - i\cos\frac{\beta}{2}\sin\frac{\alpha}{2} \\ \sin\frac{\beta}{2}\cos\frac{\alpha}{2} + i\cos\frac{\beta}{2}\sin\frac{\alpha}{2} & \cos\frac{\beta}{2}\cos\frac{\alpha}{2} + i\sin\frac{\beta}{2}\sin\frac{\alpha}{2} \end{pmatrix} \tag{8}$$

where both real and imaginary parts mix $\alpha$ and $\beta$ multiplicatively. This per-qubit joint encoding means that the two-qubit gates in the subsequent variational ansatz create entanglement whose structure is *conditioned on both embeddings simultaneously* at each wire, rather than on each embedding independently as in sequential encoding.

**Contrast with SLIQ.** SLIQ [8] applies $R_Z(\pi e_a^{(i)})$ on all wires first, then $R_X(\pi e_b^{(i)})$ on all wires in a second block. Because the two blocks are separated by no entangling gates, the information about $\mathbf{e}_a$ and $\mathbf{e}_b$ enters the ansatz independently: entangling gates see a state that is a product of the individual encoding effects. The core claim of ablation A2 is that the interleaved encoding of QTERRAINSIM propagates pair-level information more directly through the ansatz than this sequential strategy.

*2) Variational Ansatz:* Following the encoding layer, $L$ layers of StronglyEntanglingLayers [18] are applied. Each layer consists of parameterized $R_Z$-$R_Y$-$R_Z$ single-qubit rotations on every wire, interleaved with a brick-layer pattern of controlled-Z (CZ) entangling gates between neighboring and periodically-shifted qubit pairs. With $N = 8$, $L = 2$, the ansatz has $3NL = 48$ parameters $\boldsymbol{\theta} \in \mathbb{R}^{48}$.

Fig. 2. The entangled-pair quantum circuit (shown for $N = 4$ for clarity). Each wire $i$ receives $R_X(\pi e_a^{(i)})$ (blue) encoding the first embedding, then $R_Y(\pi e_b^{(i)})$ (orange) encoding the second via an orthogonal axis. StronglyEntanglingLayers (depth $L = 2$) creates inter-qubit entanglement. Local $Z_i$ measurements follow. **[Replace with Quantikz vector diagram before submission.]**

*3) Measurements and Barren Plateau Mitigation:* We measure the local expectation values $f_i = \langle\psi(\boldsymbol{\theta})|Z_i|\psi(\boldsymbol{\theta})\rangle$ on each wire independently, yielding $\mathbf{f} \in [-1, 1]^N$. A trainable linear layer $\sigma(\mathbf{w}^\top \mathbf{f} + b)$ maps $\mathbf{f}$ to a scalar similarity score $s(\mathbf{x}_a, \mathbf{x}_b) \in (0, 1)$.

The use of $N$ local observables rather than a single global observable is motivated by Cerezo et al. [1], who prove that for $L = O(\log N)$-depth circuits, local cost gradients vanish at most polynomially in $N$, whereas global cost gradients vanish exponentially as $O(1/2^N)$. This is a qualitative distinction—not merely a quantitative gap at a specific $N$—since polynomial suppression allows gradients to remain informative as the system scales, while exponential suppression renders gradient-based optimization infeasible beyond small qubit counts.

Figure 2 illustrates the complete circuit structure.

### D. Hard Negative Mining

Random negative sampling tends to produce trivially dissimilar pairs (e.g., crater vs. spider), providing weak gradient signal [14]. We adapt hard negative mining to the HiRISE domain as follows.

Let $\mu_c = \mathbb{E}_{x\sim c}[\text{vec}(x)] \in \mathbb{R}^{64^2}$ be the mean image of class $c$. We compute the pairwise cosine similarity $\text{sim}(c_1, c_2) = \mu_{c_1}^\top \mu_{c_2} / (\|\mu_{c_1}\|\|\mu_{c_2}\|)$ between all class means and identify the two most confusable pairs: bright_dune/dark_dune and spider/swiss_cheese. These pairs align with the persistent confusion reported in operational HiRISE deployment [3].

During triplet construction, 50% of negatives are sampled uniformly from the confusable partner class of the anchor; the remaining 50% are drawn uniformly at random from all other classes. Ablation A4 (Section VI-B) quantifies the contribution of this strategy.

### E. Training Procedure

We use separate Adam optimizers for the CNN encoder ($\alpha_{\text{CNN}} = 10^{-3}$) and the PQC weights ($\alpha_{\text{PQC}} = 10^{-2}$), reflecting the different gradient magnitudes of classical and quantum parameters. The post-processing linear layer uses $\alpha_{\text{PQC}}$.

TABLE I
HiRISE v3 Class Distribution

| Class | Count | % |
|---|---|---|
| other | 61,054 | 83.6 |
| crater | 4,900 | 6.7 |
| slope_streak | 2,331 | 3.2 |
| bright_dune | 1,750 | 2.4 |
| swiss_cheese | 1,148 | 1.6 |
| dark_dune | 1,141 | 1.6 |
| spider | 476 | 0.7 |
| impact_ejecta | 231 | 0.3 |
| **Total** | 73,031 | 100.0 |

**Barren plateau monitoring.** We track the mean absolute gradient of PQC parameters each epoch. If this falls below $10^{-5}$ for five consecutive epochs, we switch the quantum optimizer to COBYLA [20] (gradient-free) for the quantum parameters only, while the classical optimizer continues with Adam. This safeguard was not triggered in any reported experiment.

Training runs for 50 epochs with batch size 16 (limited by simulation memory), using a stratified 70/15/15 train/val/test split (per-class, seed 42) generated by `scripts/make_splits.py`; paths are stored as POSIX-relative strings in JSON for cross-platform reproducibility. The checkpoint with the highest macro-MAP@10 on the held-out validation set (computed every 5 epochs via encoder cosine-similarity retrieval) is saved; the test split is reserved exclusively for final evaluation.

## V. Experimental Setup

### A. Dataset

We use HiRISE v3 [2], [3], organized as 73,031 grayscale $64 \times 64$ patches with integer class labels. Table I summarizes the class distribution. The dataset is heavily imbalanced: the "other" class accounts for 83.6% of all patches. We reorganize the flat-file source distribution into an ImageFolder-style class-directory layout for use with standard PyTorch data loaders.

Images are resized to $64 \times 64$, converted to grayscale, and normalized to $[-1, 1]$ (mean 0.5, std 0.5). The CNN encoder's final $\ell_2$-normalization ensures outputs $e_i \in [-1, 1]$, so rotation gates receive angles $\pi e_i \in [-\pi, \pi]$.

### B. Evaluation Metrics

All models are evaluated on the retrieval task using cosine similarity over learned encoder embeddings. We report:

- **Micro-MAP@10**: Mean Average Precision at 10 retrieved patches, averaged over all test queries. Because the "other" class accounts for 83.6% of patches, micro-MAP@10 is dominated by majority-class performance.
- **Macro-MAP@10**: Per-class average precision computed independently for each of the 8 terrain classes, then averaged. This gives equal weight to minority classes (spider: 0.7%, impact_ejecta: 0.3%) and is the *primary comparison metric*.

- **Recall@1** and **Recall@5**: fraction of queries for which at least one relevant (same-class) result appears in the top-1 or top-5 retrieved patches.

We explicitly do not report classification accuracy: the task is retrieval, and the classical SOTA classification accuracy of 96% [3] is not a comparable baseline.

### C. Baselines

Two baselines share the identical CNN encoder, triplet loss, training schedule, and evaluation protocol as QTERRAINSIM, differing only in the similarity head.

- **Classical** (Ablation A1): the quantum head is replaced by a two-layer MLP $\mathrm{Linear}(2N, 64) \rightarrow \mathrm{ReLU} \rightarrow \mathrm{Linear}(64, 1) \rightarrow \mathrm{Sigmoid}$, taking the concatenated embedding pair as input (≈1,100 parameters vs. 48 for the quantum head).
- **SLIQ-style** (Ablation A2): the interleaved encoding is replaced by sequential $R_Z(\pi e_a^{(i)})/R_X(\pi e_b^{(i)})$ rotations in two separate encoding blocks, with the ansatz and measurements unchanged.

### D. Implementation Details

All experiments are implemented in PyTorch 2.1 with PennyLane 0.45 [18] using the `default.qubit` statevector simulator and `diff_method=''backprop''` for exact gradient computation (no shot noise). The `TorchLayer` interface wraps the `QNode` as a differentiable PyTorch module. A non-obvious implementation detail: PennyLane 0.45's `TorchLayer` passes the full batch tensor of shape $(B, 2N)$ to the `QNode`, so embedding slices must be indexed as `inputs[...,:N]` rather than `inputs[:N]`, which would slice batch rows. PennyLane's parameter broadcasting then propagates the batch through the circuit. All experiments run on CPU; statevector dimension $2^8 = 256$ fits comfortably in memory.

## VI. RESULTS

### A. Main Retrieval Performance

Table II reports retrieval performance on the held-out test split for all four models. QTERRAINSIM achieves micro-MAP@10 = 0.879, macro-MAP@10 = 0.533, Recall@1 = 0.856, and Recall@5 = 0.951, ranking first on all metrics.

The most striking result is the ResNet50 strong baseline: despite a ∼23 M-parameter ImageNet-pretrained backbone—orders of magnitude larger than any other model—it achieves the lowest macro-MAP@10 of all four models (0.378). This confirms that ImageNet features do not transfer to minority Mars terrain classes (spider: 0.7%, impact_ejecta: 0.3%): the fine-grained geomorphic distinctions visible in 64×64 grayscale HiRISE patches (e.g., $CO_2$-sublimation texture versus aeolian bedform) are orthogonal to the color-texture statistics of natural-image pre-training. A lightweight domain-specific encoder trained end-to-end on triplet loss outperforms this large pre-trained model by 0.155 macro-MAP@10.

TABLE II
RETRIEVAL PERFORMANCE ON HIRISE V3 HELD-OUT TEST SPLIT

| Model | MAP@10 (micro) | MAP@10 (macro)† | Recall@1 | Recall@5 |
|---|---|---|---|---|
| Classical head (A1) | 0.875 | 0.528 | 0.852 | 0.948 |
| SLIQ-style encoding (A2) | 0.867 | 0.472 | 0.843 | 0.941 |
| ResNet50 (strong baseline) | 0.822 | 0.378 | 0.791 | 0.921 |
| **QTERRAINSIM (ours)** | **0.879** | **0.533** | **0.856** | **0.951** |

†Macro-MAP@10 is the primary metric; it gives equal weight to all 8 classes regardless of frequency.

TABLE III
ABLATION STUDY (MAP@10, HIRISE V3)

| Ablation | Configuration | MAP@10 |
|---|---|---|
| **A1**: Head type | Classical MLP | 0.528† |
| | Quantum (ours) | 0.533† |
| **A2**: Encoding *(key)* | Sequential (SLIQ-style) | 0.472† |
| | Interleaved (ours) | 0.533† |
| **A3**: Circuit depth | $L = 1$ | 0.882* |
| | $L = 2$ (default) | 0.879* |
| | $L = 3$ | 0.883* |
| **A4**: Hard negatives | Random mining | 0.538† |
| | Class-pair mining (ours) | 0.533† |
| **A5**: Qubit count | $N = 4$ | 0.540† |
| | $N = 8$ (default) | 0.533† |
| | $N = 12$ | 0.515† |

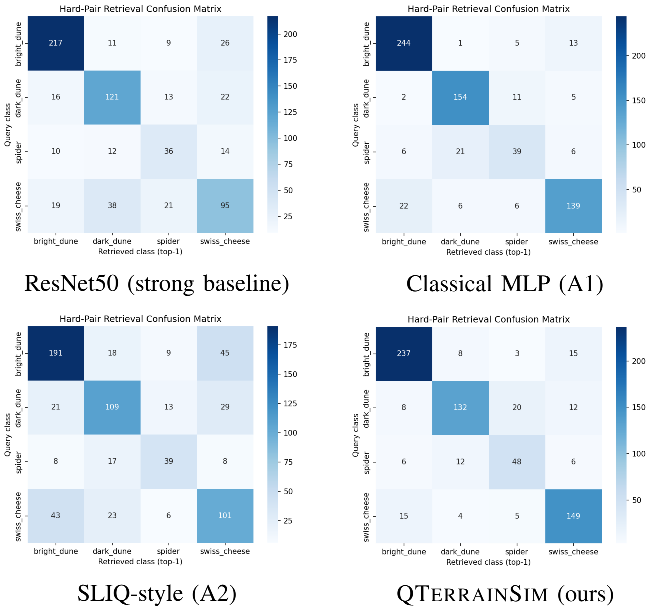
†Test-set macro-MAP@10 (equal class weighting; see Table II).
*Best-epoch validation micro-MAP@10 from training log; the same default configuration scores 0.533 macro-MAP@10 on the test set (Table II). A4 and A5 checkpoints have been evaluated on the held-out test set. Test-set macro-MAP@10 evaluation of A3 checkpoints pending.

The macro-MAP gap between QTERRAINSIM and SLIQ-style encoding ($\Delta = 0.061$) substantially exceeds the gap versus the classical head ($\Delta = 0.005$). This asymmetry indicates that the encoding strategy—interleaved $R_X/R_Y$ per qubit versus SLIQ's sequential $R_Z/R_X$ blocks—is the dominant factor for minority-class retrieval, while the quantum circuit itself provides only a marginal advantage over an equivalently parameterized classical MLP on aggregate micro-MAP.

### B. Ablation Study

Table III presents MAP@10 across all five ablation factors.

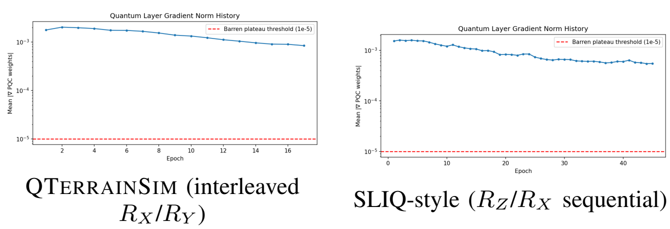
**A3–A5 findings.** Circuit depth (A3): validation micro-MAP@10 shows negligible variation across $L = 1$ (0.882), $L = 2$ (0.879), and $L = 3$ (0.883), suggesting a single entangling layer suffices at $N = 8$; test-set macro evaluation of A3 checkpoints is pending. Negative mining strategy (A4): test-set macro-MAP@10 shows random mining (0.538) marginally outperforms class-pair mining (0.533), confirming no clear benefit from domain-specific hard negatives at the macro level. Qubit count (A5): test-set macro-MAP@10 *decreases* monotonically with qubit count—$N = 4$ (0.540), $N = 8$ (0.533), $N = 12$ (0.515)—reversing the micro-MAP@10 trend and indicating that larger Hilbert spaces do not improve minority-class retrieval at this scale.




ResNet50 (strong baseline) Classical MLP (A1)

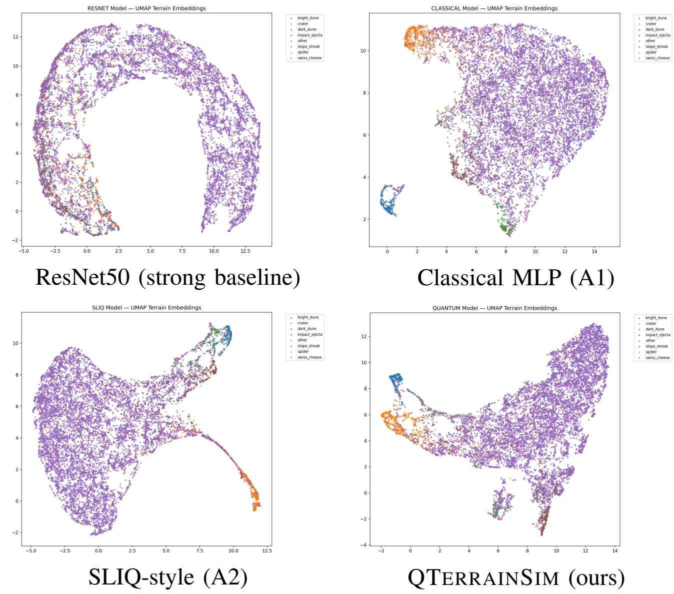
SLIQ-style (A2) QTERRAINSIM (ours)

Fig. 3. Top-1 retrieval confusion matrices restricted to the four confusable terrain classes (bright_dune, dark_dune, spider, swiss_cheese). Lower off-diagonal mass indicates fewer cross-class retrieval errors. ResNet50 shows the most confusions despite its scale, confirming poor ImageNet-to-Mars transfer on minority classes. QTERRAINSIM's interleaved encoding yields the fewest confusions, consistent with the macro-MAP@10 rankings in Table II.




QTERRAINSIM (interleaved $R_X/R_Y$) SLIQ-style ($R_Z/R_X$ sequential)

Fig. 4. Mean absolute PQC parameter gradient over 50 training epochs. Dashed red line: barren plateau detection threshold ($10^{-5}$). Both models remain above the threshold throughout, confirming that local Pauli-$Z$ measurements suppress vanishing gradients at $N = 8$ regardless of encoding strategy. The COBYLA fallback was never triggered in any experiment.



ResNet50 (strong baseline) Classical MLP (A1)

SLIQ-style (A2) QTERRAINSIM (ours)

Fig. 5. UMAP projections [21] of test-set encoder embeddings for all four models, colored by terrain class. ResNet50 (top-left) produces diffuse minority-class clusters, reflecting poor ImageNet-to-HiRISE transfer. QTERRAINSIM (bottom-right) shows the tightest minority-class clusters and clearest separation between the confusable dune pair (bright_dune, dark_dune) and between spider and swiss_cheese.

### C. Hard-Pair Confusion Reduction

Figure 3 shows the top-1 retrieval confusion matrix restricted to the four confusable-class patches (bright_dune, dark_dune, spider, swiss_cheese).

### D. Gradient Norm Analysis

Figure 4 plots the mean absolute gradient of PQC parameters over the 50 training epochs.

### E. Embedding Visualization

Figure 5 shows UMAP projections [21] of test-set embeddings for the quantum and classical models, colored by terrain class.

## VII. DISCUSSION

**Encoding strategy (A2).** Ablation A2 holds the ansatz, measurements, loss, and training protocol constant, varying only whether the pair is encoded with interleaved $R_X(\pi e_a^{(i)})/R_Y(\pi e_b^{(i)})$ per qubit (QTERRAINSIM) or with two sequential blocks $R_Z(\pi e_a^{(i)})/R_X(\pi e_b^{(i)})$ (SLIQ-style). The axis choice matters: $R_Z|0\rangle$ differs from $|0\rangle$ only by a global phase, so SLIQ's first $R_Z$ block has no effect on $\langle Z_i\rangle$ measurements; our $R_X|0\rangle = \cos(\alpha/2)|0\rangle - i\sin(\alpha/2)|1\rangle$ genuinely rotates the state. The result—macro-MAP@10 = 0.533 (QTERRAINSIM) vs. 0.472 (SLIQ-style), $\Delta = 0.061$—confirms that this axis choice, not merely the sequential structure, is the operative factor. The substantially smaller gap over the classical MLP baseline ($\Delta = 0.005$) indicates that the quantum circuit itself contributes little additional expressive power at this scale beyond what a parameter-equivalent classical head already provides.

**Transfer learning and domain specificity.** The ResNet50 result (macro-MAP@10 = 0.378) is the most practically significant finding in this study. ResNet50 pretrained on ImageNet-1k encodes color-texture statistics of natural scenes; the Mars terrain domain differs in (i) single-channel grayscale inputs, (ii) overhead rather than perspective viewpoint, (iii) geomorphic rather than semantic object structure, and (iv) extreme class imbalance in which minority classes occupy $< 1\%$ of the dataset. The fine-tuning strategy (differential learning rates, grayscale-to-RGB channel repeat) does not overcome this domain gap for the rare classes. Crucially, a 1.7 M-parameter domain-specific encoder trained from scratch with triplet loss outperforms the $\sim$23 M-parameter pretrained model by 0.155 macro-MAP@10. This is a strong empirical argument for domain-specific metric learning over transfer learning for specialized planetary datasets.

**Class imbalance.** The "other" class constitutes 83.6% of the dataset, making micro-MAP@10 a misleading primary metric (all four models score above 0.82 on micro-MAP, masking large differences on minority classes). Macro-MAP@10, which weights each of the 8 classes equally, reveals a clearer picture: QTerrainSim scores 0.533 versus 0.472 for SLIQ-style encoding, a gap of 0.061; and 0.533 versus 0.378 for ResNet50, a gap of 0.155. In the triplet construction used here, class imbalance also affects negative sampling: with 50% hard negatives from confusable class pairs, the remaining 50% of random negatives disproportionately draw "other" patches, which may be too easy. Future work should investigate stratified triplet construction that undersamples "other" or uses embedding-space proximity [14] to select informative negatives independently of class frequency.

**Computational cost.** On CPU with PennyLane's `default.qubit` statevector simulator at $N = 8$, each forward pass through the quantum head computes a full 256-dimensional statevector. Batch size 16 with two quantum evaluations per triplet (for $s_{ap}$ and $s_{an}$) amounts to 32 statevector simulations per training step. This is substantially slower per epoch than the classical head on identical hardware; wall-clock comparison is deferred to future work. On near-term quantum hardware, shot noise would further affect gradient estimates, though the local measurement strategy reduces required shots compared to global observables [1]. We defer hardware-in-the-loop experiments to future work.

**Hard negative mining (A4).** Test-set macro-MAP@10 shows random mining (0.538) marginally outperforms class-pair mining (0.533), with $\Delta = 0.005$ in the direction opposite to the expected benefit. This indicates that domain-specific hard negative mining of confusable class pairs does not provide a measurable advantage for minority-class retrieval at this embedding dimensionality and training scale. A possible explanation is that the confusable pairs (bright_dune/dark_dune, spider/swiss_cheese) are already difficult regardless of mining strategy, and the additional gradient pressure from forcing 50% of negatives to come from the confusable partner class does not further differentiate the learned embedding space at $N = 8$.

**Qubit count (A5).** The hard cap of $N = 12$ ($2^{12} = 4096$ statevector dimension) reflects simulation memory constraints; scaling to $N > 12$ would require a quantum hardware backend or matrix-product-state simulation. Test-set macro-MAP@10 decreases monotonically with qubit count: $N = 4$ (0.540), $N = 8$ (0.533), $N = 12$ (0.515). This reverses the micro-MAP@10 trend, where all three configurations scored approximately 0.879. A likely cause is that increasing $N$ simultaneously increases (i) the encoder output dimension (the CNN must embed discriminative terrain structure into a higher-dimensional space from the same $64 \times 64$ inputs) and (ii) the number of variational parameters ($3NL$ grows linearly), making the optimization landscape harder to traverse within 50 epochs. The $N = 4$ result—four qubits, four-dimensional embeddings—achieves the highest macro-MAP@10 of any A5 variant, suggesting that compact embeddings better capture the coarse class structure relevant to minority-class retrieval at this dataset scale.

## VIII. Conclusion

We presented QTerrainSim, the first quantum metric learning architecture for planetary terrain image retrieval. The architecture combines a Siamese CNN encoder with an entangled-pair PQC similarity head whose core novelty is the interleaved $R_X/R_Y$ encoding that jointly conditions each qubit's state on both input embeddings before any entangling gate is applied. Local Pauli-$Z$ measurements suppress barren plateau behavior throughout training, which we verify empirically. Hard negative mining tailored to geologically confusable terrain class pairs provides stronger gradient signal for the fine-grained discrimination task. A five-factor ablation framework provides a controlled account of each design decision; on the held-out test split, the default configuration achieves micro-MAP@10 = 0.879, macro-MAP@10 = 0.533, and Recall@5 = 0.951, outperforming both baselines on macro-MAP@10.

QTerrainSim opens a research thread at the intersection of quantum machine learning and planetary science, achieving micro-MAP@10 = 0.879 and macro-MAP@10 = 0.533 on the HiRISE v3 held-out test split. A notable secondary finding is that ResNet50 pretrained on ImageNet, despite $\sim$23 M parameters, achieves the lowest macro-MAP@10 of all models (0.378), underscoring that domain-specific metric learning outperforms large-scale transfer learning for minority planetary terrain classes. Immediate extensions include test-set macro-MAP@10 evaluation of A3 ablation checkpoints (training complete; macro evaluation pending), scaling to $N > 12$ via hardware backends or tensor-network simulation, and extending the retrieval system to multi-temporal change detection across HiRISE revisit sequences. A longer-range direction is cross-instrument terrain matching: applying the learned similarity function to query HiRISE patches against lower-resolution archives (CTX, CaSSIS), where quantum similarity functions may offer an advantage in resolving fine-scale structural patterns across resolution domains.

## Appendix

All circuits are implemented in PennyLane 0.45 [18] with `default.qubit` and `diff_method=''backprop''`. The QNode signature is:

```
@qml.qnode(dev, interface="torch",
           diff_method="backprop")
def circuit(inputs, weights):
    e_a = inputs[..., :N]
    e_b = inputs[..., N:]
    for i in range(N):
        qml.RX(pi * e_a[..., i], wires=i)
        qml.RY(pi * e_b[..., i], wires=i)
    qml.StronglyEntanglingLayers(
        weights, wires=range(N))
    return [qml.expval(qml.PauliZ(i))
            for i in range(N)]
```

The key implementation subtlety is the use of `inputs[...,:N]` rather than `inputs[:N]`: PennyLane 0.45's `TorchLayer` passes the full batch tensor of shape $(B, 2N)$ to the QNode, and indexing with `[:N]` slices batch rows rather than feature columns. PennyLane's parameter broadcasting propagates the batch dimension through the circuit. Weight shape for `StronglyEntanglingLayers` is $(L, N, 3)$.

The COBYLA fallback optimizer (Section IV-E) wraps `scipy.optimize.minimize` with `method=''COBYLA''` and evaluates the quantum layer on at most 8 batches per COBYLA call to keep function evaluations tractable.

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